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ThingBoot IoT Platform & Device Local-Deployment Solution

Efficient, secure, scalable IoT access platform



Platform Architecture

Core Functions

Working Principle

Local Deployment

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01 Platform Architecture

Robust, extendable IoT platform, create seamless intelligent connecting eco-system



Platform Architecture Diagram



01 Business Layer

IoT Console Core

Unified hub for device control, product management, open platform, private deployment, mall, CRM, CMS, payment gateway, and permission control systems.

Multi-Client Access

Seamless experience across Mobile APP, WeChat Mini Program, and PC Web Portal, ensuring consistent user interaction across all touchpoints.



02 Facility Layer

Message Server Suite

Core engine featuring state machine, OTA firmware updates, NTP sync, log/event services, debugging tools, and configurable command/cache components.

High-Performance Broker

Lightweight MQTT broker enabling reliable, low-latency device-to-cloud communication for real-time data streaming and command delivery.



03 Infrastructure Layer

Server Cluster Backbone

Enterprise-grade cluster with log analysis, big data engine, distributed caching, message queuing, disaster recovery, and resource monitoring.

Flexible Deployment

Hybrid architecture supporting public cloud scalability, private cloud security, and local on-premises deployment for data sovereignty.

End-to-End IoT Architecture: Delivering scalable, secure, and flexible solutions from user interaction to robust infrastructure.

Facility-Layer System Components



🌐 **Log Service:** Efficiently ingests, processes, and stores logs from all connected devices, providing a comprehensive audit trail for monitoring and troubleshooting.

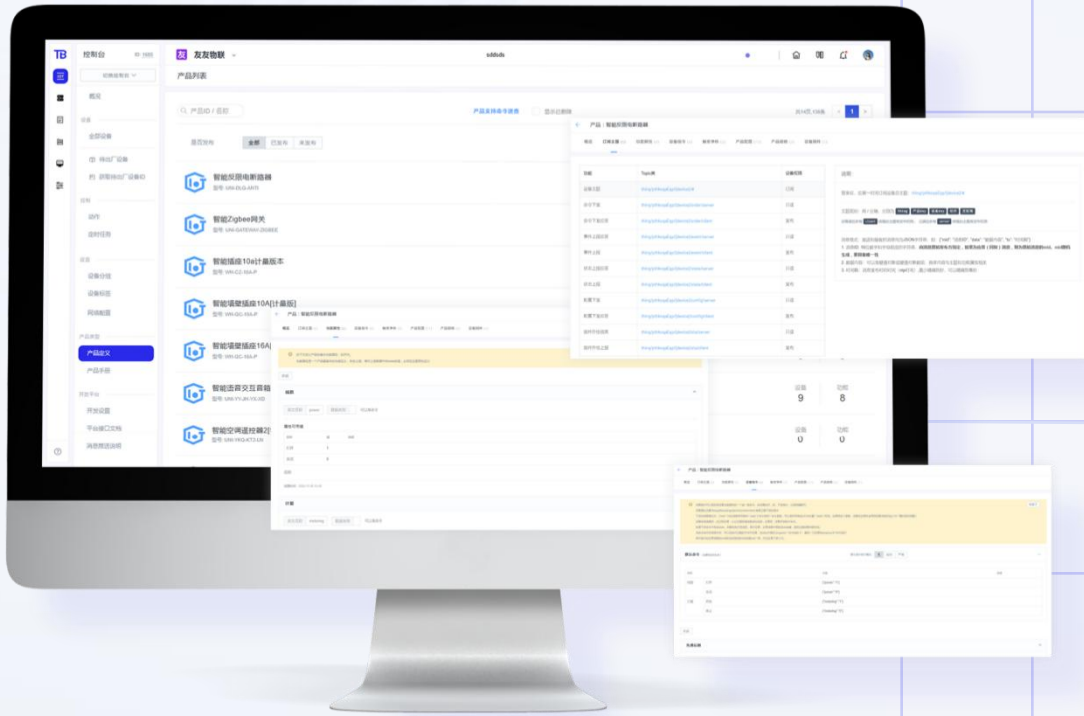
🌐 **Data Engine:** Powers real-time data processing and advanced analytics, transforming raw device telemetry into actionable insights for operational optimization.

🌐 **Cache & Queue Mechanism**

Ensures high throughput and reliable message delivery under heavy load; augmented by Disaster Recovery & Resource Analysis for maximum system stability.



Infrastructure-Layer Server Cluster



Cloud
Deployment

On-premises

Msg & Broker

OTA & NTP

Log Service

API & Debug

Config &
Cache

Event & Cmd

High-Performance Core

A robust cluster foundation supporting elastic scalability and full control over your IoT infrastructure, with essential services for seamless device management.

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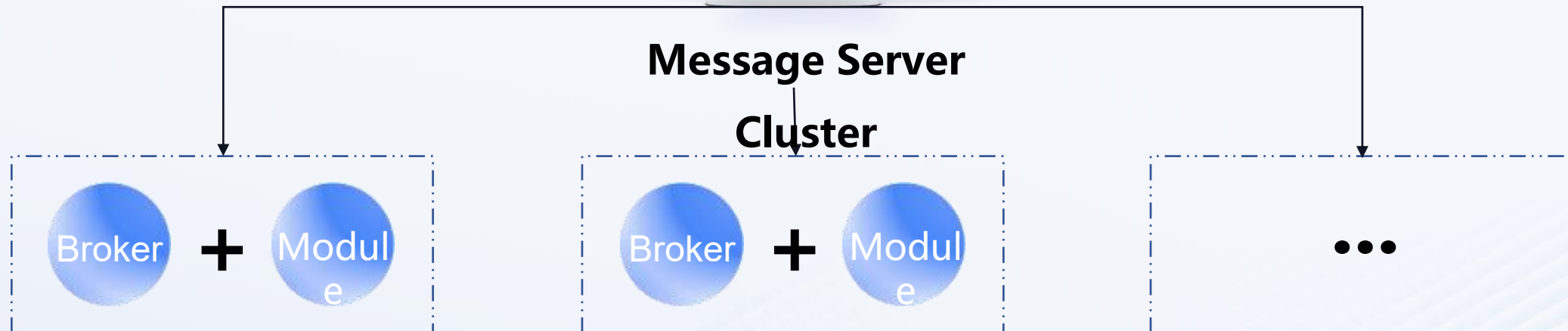
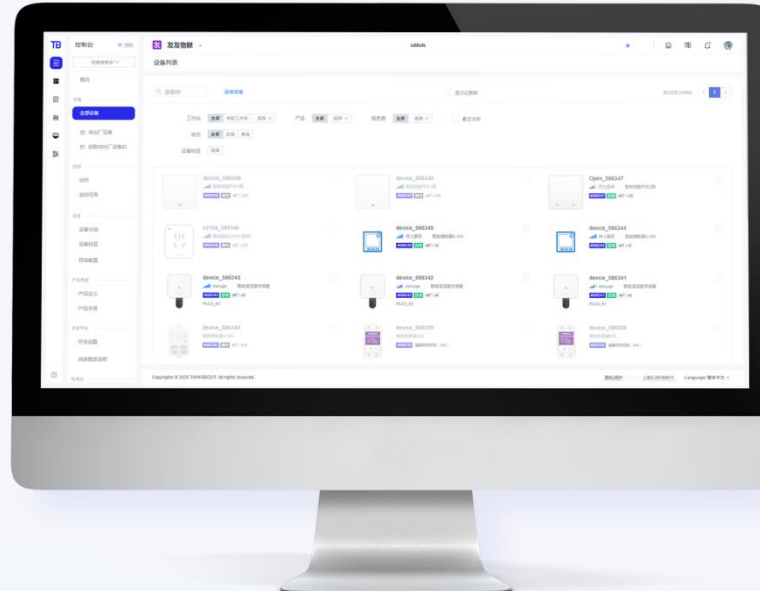
02 Core Functions

Implement flexible & efficient IoT device management, and controllable core engine



Core Function - IoT Console

An IoT-device management console/portal..

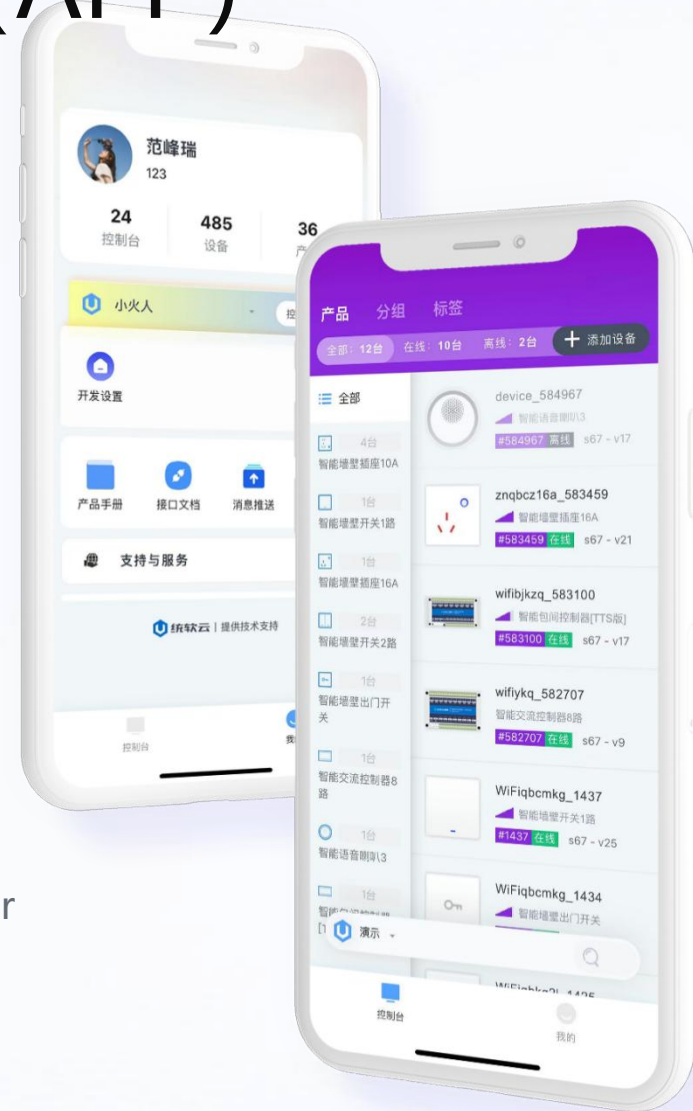


It connects and manages all devices via the message-server cluster and provides a unified entry for device management

Core Functions - User Client (APP)

- White-box APP: The UI and functions support in-depth customization to align with brand positioning
- Store Module: Supports integration of online malls to expand equipment sales

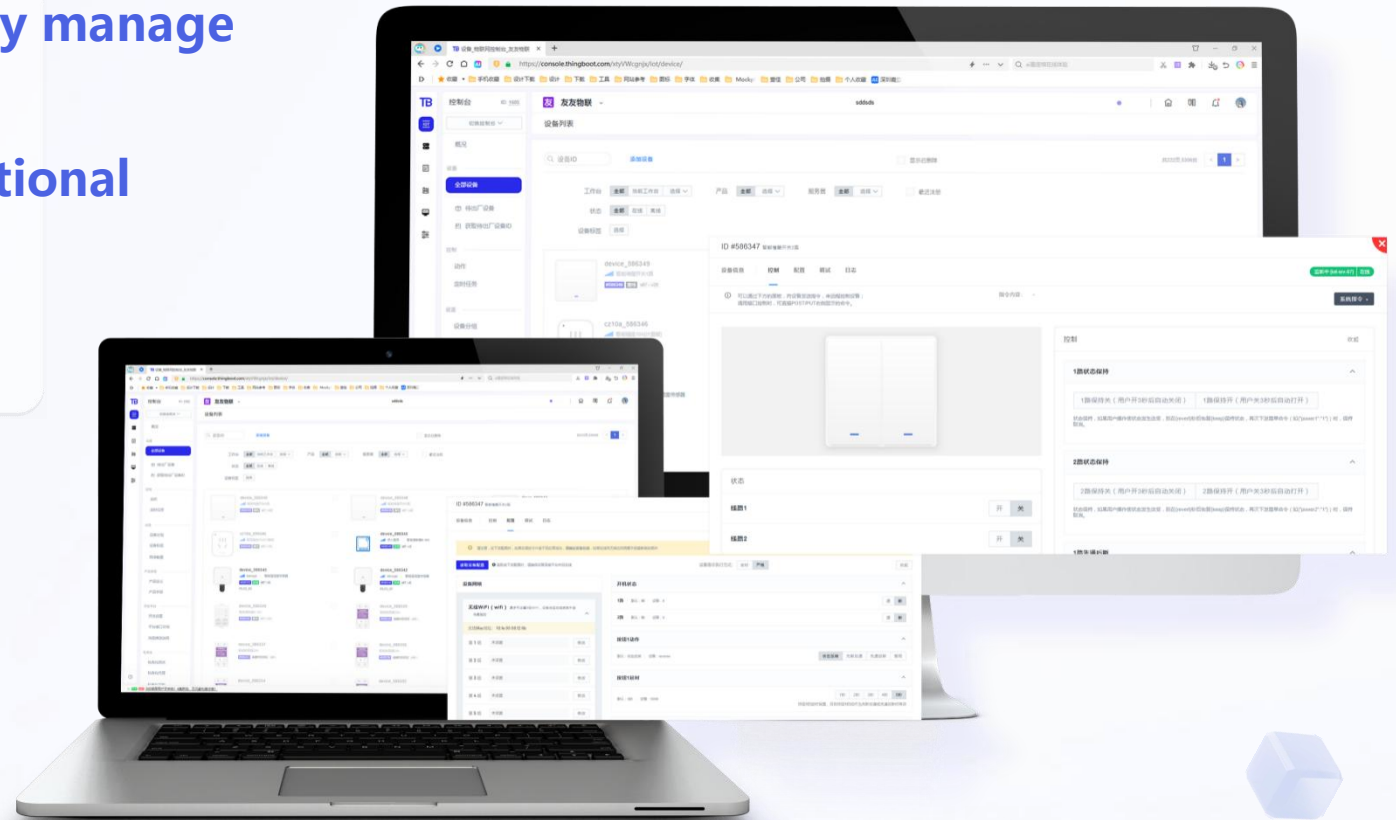
NOTE: Provide end users with convenient device management and interactive experience, and build a one-stop intelligent device control center



Core Functions - User Client Side (Web Portal)

- Access via web browser: Directly manage and control devices
- Store Module: Selectable & Optional integration of online mall

NOTE: Supports cross-platform device management capabilities, breaks down the boundaries between devices and systems, and delivers collaborative multi-terminal control and a consistent user experience.



Core Functions - Open Platform



- Fine-grained Permission Management: Allocate device control permissions by workspace units to achieve refined permission governance.
- Comprehensive API Access: Full device management interfaces are provided to enable seamless integration and connection with third-party systems.
- Flexible Ecosystem Integration: Supports two-way interaction between third-party systems and devices to jointly build an open intelligent ecosystem.

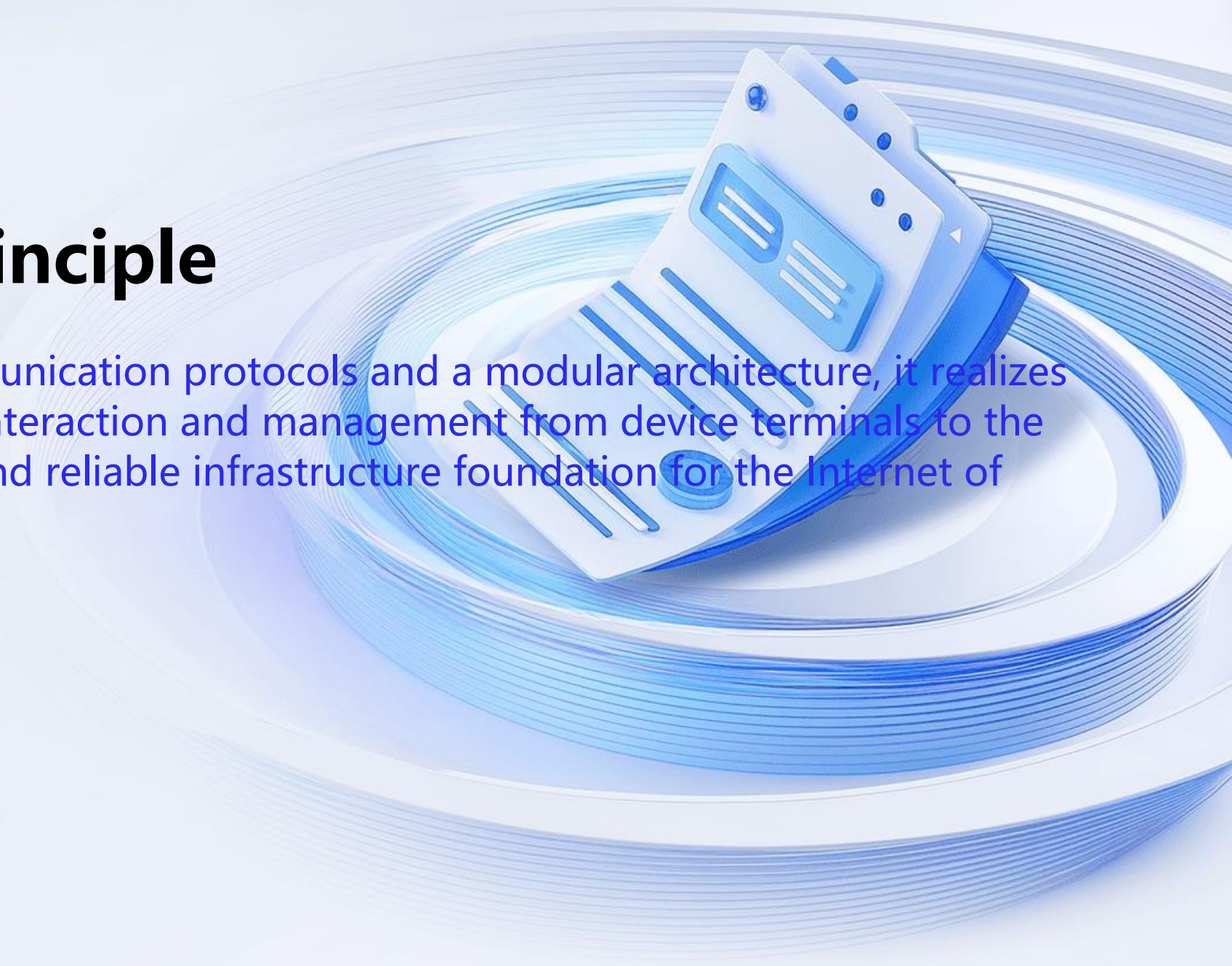
Empower cross-platform device control and data interconnection, break down system barriers, and expand business application scenarios.



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03 Working Principle

Based on lightweight communication protocols and a modular architecture, it realizes efficient, low-latency data interaction and management from device terminals to the cloud, and builds a stable and reliable infrastructure foundation for the Internet of Things.



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04 Local Deployment

Consolidate data sovereignty & Compliant safety defense borderline

Built exclusively for enterprises with stringent requirements for data privacy and security compliance, it supports private deployment on enterprises' own servers or on-premises computer rooms to achieve full independent control over data storage and access. This solution fully complies with compliance standards of highly regulated industries such as finance, government affairs and healthcare. While safeguarding data security, it guarantees high availability of business systems and localized operation and maintenance capabilities.



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05 Contact Us

On demand technical support, together to catch new opportunity



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